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StructSynth: Dependency Graphs as Generation Plans for Low-Data Tabular Synthesis with Language Models

 (/pdf?id=32AX5rq8ip)

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 26 May 2026 (modified: 24 Jun 2026)  ACL ARR 2026 May Submission
 May, Senior Area Chairs, Area Chairs, Reviewers, Authors, Secondary Reviewers
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Keywords: Tabular Data Synthesis, Dependency Structure Discovery, Structure- Conditioned Generation, Data Augmentation, Low-Data Regimes

Abstract:

Tabular data derives its value from inter-feature dependencies, yet preserving them during synthesis is fragile when samples are scarce. Existing approaches either learn dependencies implicitly through distribution fitting, rely on statistical graph learning that becomes unstable with few samples, or encode structure through flat text serialization. Recent graph-aware methods use dependency graphs as attention biases or as backbones for non-LLM samplers, but they do not use the graph as a prompt-level plan for organizing the LLM's own generation process. We introduce StructSynth, a framework that treats a dependency graph as a generation plan---determining the generation order, conditioning context, and scope of each black-box LLM call. In Evidence-Grounded Graph Induction, LLM reasoning and statistical association cues jointly construct a Directed Acyclic Graph (DAG) from limited samples. In Graph-Planned Conditional Synthesis, this DAG drives autoregressive synthesis in topological order, conditioning each feature on previously generated values with graph-specified parent structure guiding each step, making the conditioning schedule explicit throughout synthesis. Experiments show that StructSynth achieves state-of-the-art downstream utility and privacy preservation while maintaining competitive statistical fidelity, especially in low-data scenarios.

Paper Type: Long

Research Area: NLP Applications

Research Area Keywords: financial/business NLP, security/privacy, NLP for social good

Contribution Types: NLP engineering experiment

Languages Studied: English

Reassignment Request Area Chair: This is not a resubmission

Reassignment Request Reviewers: This is not a resubmission

A1 Limitations Section: This paper has a limitations section.

A2 Potential Risks: Yes

A2 Elaboration: In Ethical Considerations section

B Use Or Create Scientific Artifacts: Yes

B1 Cite Creators Of Artifacts: Yes

B1 Elaboration: Section 4 and Appendix A cite the public datasets, baselines, and software libraries used in the experiments.

B2 Discuss The License For Artifacts: No

B2 Elaboration: We cite the artifacts used, but we do not provide a detailed license-by-license discussion. The artifacts are used only for research benchmarking.

B3 Artifact Use Consistent With Intended Use: Yes

2026/7/8 21:17StructSynth: Dependency Graphs as Generation Plans for Low-Data Tabular Synthesis with Language Models | OpenReview

B3 Elaboration: The Ethical Considerations section states that public datasets are used only as research benchmarks, and that generated records should not be used for high-stakes decision making without additional review and validation.

B4 Data Contains Personally Identifying Info Or Offensive Content: No

B4 Elaboration: We use publicly available benchmark datasets and do not collect new data. We discuss data sensitivity and privacy risks in Ethical Considerations, but we do not provide a separate PII/offensive-content audit.

B5 Documentation Of Artifacts: Yes

B5 Elaboration: Section 4 and Appendix "Detailed Experimental Setup" document the datasets, including domains, tasks, feature types, and evaluation setup.

B6 Statistics For Data: Yes

B6 Elaboration: Section 4 and Appendix "Detailed Experimental Setup" report dataset statistics, train/test splits, few-shot sample sizes, and random seeds.

C Computational Experiments: Yes

C1 Model Size And Budget: Yes

C1 Elaboration: Appendix "Implementation and Hyperparameters" reports model settings and downstream model parameters. Appendix "Efficiency Analysis" reports token usage for CLLM and StructSynth, including input/output tokens, total tokens, and relative overhead. Since the main LLMs are hosted/API models, we report token budget rather than GPU hours.

C2 Experimental Setup And Hyperparameters: Yes

C2 Elaboration: Section 4 and Appendix "Detailed Experimental Setup" describe the experimental setup, including datasets, baselines, train/test splits, sample sizes, seeds, and hyperparameter settings.

C3 Descriptive Statistics: Yes

C3 Elaboration: Section 4 reports results over multiple datasets and random seeds, including mean performance and variability/statistical comparisons where applicable.

C4 Parameters For Packages: Yes

C4 Elaboration: Appendix "Detailed Experimental Setup" reports implementation details and parameter settings for baselines, downstream models, LLMs, and evaluation components.

D Human Subjects Including Annotators: No

D1 Instructions Given To Participants: N/A

D2 Recruitment And Payment: N/A

D3 Data Consent: N/A

D4 Ethics Review Board Approval: N/A

E Ai Assistants In Research Or Writing: Yes

E1 Information About Use Of Ai Assistants: No

E1 Elaboration: We used AI assistants only for language polishing and editing support. All technical content, experiments, analyses, claims, and final writing decisions were reviewed and approved by the authors.

Author Submission Checklist: yes

EMNLP 2026 AI Reviewing Experiment: yes

TLDR: StructSynth learns a dataset's hidden structure and uses it as a blueprint to guide an LLM, generating more realistic synthetic data from scarce samples.

Preprint: no

Preprint Status: There is a non-anonymous preprint (URL specified in the next question).

Existing Preprints: <https://arxiv.org/abs/2508.02601> (<https://arxiv.org/abs/2508.02601>)

Preferred Venue: EMNLP

Visa Needs: yes

Consent To Share Data: yes

Consent To Share Submission Details: On behalf of all authors, we agree to the terms above to share our submission details.

Association For Computational Linguistics - Blind Submission License Agreement: On behalf of all authors, I agree

Submission Number: 9229

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Official Review of Submission9229 by Reviewer 7Wh2

Official Review by Reviewer 7Wh2  06 Jul 2026, 09:53 (modified: 08 Jul 2026, 20:34)

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 7Wh2

 Revisions (/revisions?id=6SiIIYQEKB)

Paper Summary:

This paper focuses on tabular data synthesis using large language models (LLMs). The idea is to first induce a dependency DAG from a small real table using both LLM reasoning and statistical association measures, then use that DAG as a generation plan for black-box LLM calls. Basically the graph is used to determine feature order and prompt conditioning context.

Summary Of Strengths:

- The paper does a nice job and is akin to Bayesian network papers that separate structure discovery from conditional generation.
- The empirical results are strong. The authors demonstrate that their approach has the best downstream performacne on the datasets considered.
- The ablations are helpful. Replacing the LLM-guided graph discovery, or removing association scores, or ignoring topological order all hurt performance.

Summary Of Weaknesses:

- My main concern with this paper is that it is quite close to the paper by S. Xu et al. "Why LLMs Are Bad at Synthetic Table Generation and what to do about it" (<https://arxiv.org/abs/2406.14541> (<https://arxiv.org/abs/2406.14541>)). This paper argues that LLM tabular generation fails because autoregressive generation imposes an order over columns, and that random feature-order fine-tuning will violate functional dependencies. It proposes injecting knowledge of functional relationships. Just like the proposed paper, there is a DAG learning step and then using that DAG information to generate data. As a result, the novelty here is limited and further the experimental results need to be contrasted with this above paper and their PAFT algorithm (in fact they seem to use the same dataset(s) as well).
- Besides the key related work contrasting and comparison, the privacy evaluation feels underdeveloped and is not quite a contribution.
- The statistical fidelity results are mixed (the algorithm proposed appears middle of the pack in some of these results).
- The limitations section acknowledges that opaque schemas such as V1/V2 or genomic locus identifiers weaken the LLM semantic prior (whereas this may not be a problem for the PAFT paper which seems to be discovering the dependencies).

Comments Suggestions And Typos:

Please see above.

Confidence: 5 = Positive that my evaluation is correct. I read the paper very carefully and am familiar with related work.

Soundness: 3 = Acceptable: This study provides sufficient support for its main claims. Some minor points may need extra support or details.

Excitement: 1.5

Overall Assessment: 2 = Resubmit next cycle: I think this paper needs substantial revisions that can be completed by the next ARR cycle.

Ethical Concerns:

There are no concerns with this submission.

Needs Ethics Review: No

Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.

Datasets: 4 = Useful: I would recommend the new datasets to other researchers or developers for their ongoing work.

Software: 4 = Useful: I would recommend the new software to other researchers or developers for their ongoing work.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Publication Ethics Policy Compliance: I did not use any generative AI tools for this review

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Official Review of Submission9229 by Reviewer q74j

Official Review by Reviewer q74j 📅 29 Jun 2026, 17:05 (modified: 08 Jul 2026, 20:34)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer q74j

📄 Revisions (/revisions?id=OY4fiQT2uE)

Paper Summary:

This paper proposes a method for generating additional tabular data in scarce data scenarios. They decouple the extraction task in which a DAG is extracted from the existing tabular data using an LLM, and the generation task, where additional data is generated using the extracted DAG as guidance. They show

Summary Of Strengths:

The paper is very well-written The experimental design is sound. The method is evaluated on six downstream tasks and the extracted graph is evaluated for its quality as well. They show with extensive experiments that method is suitable for the task at hand The ablations further strengthen the claims and show the contribution of the different elements in the design

Summary Of Weaknesses:

It is not clear whether the baselines used are the best available methods in their respective categories. As the authors explain in l.052-l.055, deep generative models' performance varies across architectures and some may require many more examples to train. I expected a little explanation in the experiments section that motivates the choice for the particular models used as baselines. It is not always clear why you select a particular dataset to show results. In Section 4.5 and 4.6 why did you choose the Adult dataset and not one of the datasets that are created after the knowledge cutoff for LLMs?

Comments Suggestions And Typos:

The goal of the paper only became fully clear to me in l. 197. This description could feature earlier in the introduction.

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.

Soundness: 4.5

Excitement: 3.5

Overall Assessment: 4 = Conference: I think this paper could be accepted to an *ACL conference.

Limitations And Societal Impact:

The limitation section is insightful and to the point. The paper also discusses ethical considerations well.

Ethical Concerns:

The authors describe ethical considerations well.

Needs Ethics Review: No

Reproducibility: 1 = They would not be able to reproduce the results here no matter how hard they tried.

Datasets: 1 = No usable datasets submitted.

Software: 1 = No usable software released.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Publication Ethics Policy Compliance: I did not use any generative AI tools for this review

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Official Review of Submission9229 by Reviewer iRH3

Official Review by Reviewer iRH3 📅 24 Jun 2026, 16:38 (modified: 08 Jul 2026, 20:34)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer iRH3

📄 Revisions (/revisions?id=54FM9gJXkT)

Paper Summary:

This paper proposes StructSynth, a tabular data generation and augmentation method that first constructs a dependency graph among features using an LLM and then generates synthetic samples based on the learned graph structure. Learning dependencies among features is a central challenge in this setting and has been studied from various perspectives, including Bayesian networks, causal discovery, and statistical approaches. The key contribution of this work is to use the semantic knowledge of LLMs to infer the dependency graph, particularly in small sample size scenarios where purely data-driven approaches may struggle.

Summary Of Strengths:

1. The paper addresses an important problem in tabular data generation and augmentation, particularly under small sample size regimes where existing methods often face difficulties.
2. Experimental results are convincing, especially in terms of downstream model performance. The proposed method consistently performs well when the available sample size is small, suggesting that it may be practically useful across a range of applications.
3. The paper goes beyond reporting aggregate performance metrics and also presents examples of the learned dependency graphs. These qualitative results provide useful insights into the behavior of the method and help readers better understand the learned structures.

Summary Of Weaknesses:

1. Since the proposed method infers not only the existence of dependencies but also the directions of edges, the evaluation should assess more than statistical fidelity. While the paper reports measures such as Statistical Fidelity Error, it would also be valuable to evaluate the correctness of the learned graph structure, particularly edge directions, on datasets where ground-truth dependency structures are available.
2. The method is specifically motivated by small sample size scenarios, and the experiments focus on sample sizes up to $n = 200$. However, it is important to understand how the method behaves as more data become available. Experiments with larger sample sizes would help clarify the trade-off between the advantages of LLM-based graph construction and purely data-driven alternatives.
3. As acknowledged in the limitations section, the proposed approach exploits semantic information contained in feature names and descriptions, whereas most non-LLM baselines cannot use such information. This raises questions about the fairness of the comparison. An informative additional experiment would be to anonymize feature names (e.g., replacing them with V1, V2, ...) and evaluate how much the performance depends on semantic information provided by the feature names.

Comments Suggestions And Typos:

Please address the weaknesses discussed above.

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.

Soundness: 3.5

Excitement: 3 = Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.

Overall Assessment: 3 = Findings: I think this paper could be accepted to the Findings of the ACL.

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.

Datasets: 3 = Potentially useful: Someone might find the new datasets useful for their work.

Software: 4 = Useful: I would recommend the new software to other researchers or developers for their ongoing work.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Publication Ethics Policy Compliance: I used a privacy-preserving tool exclusively for the use case(s) approved by PEC policy, such as language edits

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Author-Editor Confidential Comment by Program Chairs

Author-Editor Confidential Comment by Program Chairs 13 Jun 2026, 14:41

Program Chairs, Senior Area Chairs, Area Chairs, Authors

Comment:

Dear authors,

In your submission you have some captions that are positioned above the tables instead of positioned below. Your submission will not be desk rejected for that, but please make sure to fix it according to the formatting guidelines at <https://acl-org.github.io/ACLPUb/formatting.html> (<https://acl-org.github.io/ACLPUb/formatting.html>) during re-submission or camera-ready stage.

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